Sampling Examples with Sample Dataset

Sample Dataset

Suppose we have a dataset of 20 students with their weekly homework hours:

Student ID	Hours per Week
1	5
2	8
3	6
4	10
5	7
6	9
7	12
8	11
9	4
10	6
11	8
12	9
13	5
14	7
15	10
16	12
17	6
18	11
19	9
20	8

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1. Random Sampling

Goal: Select 5 students randomly from the dataset.
Step-by-step:

1. Assign each student a number from 1 to 20.
2. Randomly select 5 numbers, e.g., 2, 7, 9, 14, 18.
3. The selected students and their hours:

Student ID	Hours per Week
2	8
7	12
9	4
14	7
18	11

Sample mean:

$$\bar{X}_{sample} = \frac{8+12+4+7+11}{5} = \frac{42}{5} = 8.4$$

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2. Cluster Sampling

Scenario: Suppose students are grouped by class:

- Class A: Students 1–5
- Class B: Students 6–10
- Class C: Students 11–15
- Class D: Students 16–20

Step-by-step:

1. Randomly select 2 classes, e.g., Class B and Class D.
2. Include all students from these classes.

Student ID	Hours per Week
6	9
7	12
8	11
9	4
10	6
16	12
17	6
18	11
19	9
20	8

Sample mean:

$$\bar{X}_{sample} = \frac{9+12+11+4+6+12+6+11+9+8}{10} = \frac{88}{10} = 8.8$$

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3. Stratified Sampling

Scenario: Suppose we divide students by workload category:

- Low (Hours ≤ 6): 1,3,5,9,10,13,17
- Medium (Hours 7–9): 2,6,11,12,14,19,20
- High (Hours ≥ 10): 4,7,8,15,16,18

Step-by-step:

1. Take 2 students randomly from each stratum.
2. Selected students:

Stratum	Student ID	Hours per Week
Low	1	5
Low	10	6
Medium	2	8
Medium	14	7
High	7	12
High	15	10

Weighted mean:

$$\bar{X}_{population} \approx \frac{7}{20}\cdot\frac{5+6}{2}+\frac{7}{20}\cdot\frac{8+7}{2}+\frac{6}{20}\cdot\frac{12+10}{2} = 0.35\cdot5.5+0.35\cdot7.5+0.3\cdot11 = 1.925+2.625+3.3 = 7.85$$

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4. Systematic Sampling

Goal: Select every 4th student starting from a random position between 1 and 4.
Step-by-step:

1. Random start, say 3. Step size = 4.
2. Selected students: 3, 7, 11, 15, 19

Student ID	Hours per Week
3	6
7	12
11	8
15	10
19	9

Sample mean:

$$\bar{X}_{sample} = \frac{6+12+8+10+9}{5} = \frac{45}{5} = 9$$